

Solution SET: MFML (MAKEUP)

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1 Question 1: Linear Systems

(a) Find the Echelon form and Determinant [2 Marks]

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Method 1: Standard Row Reduction

We perform Gaussian elimination to reach Row Echelon Form.

Step 1: Eliminate entries in Column 1 Target $A_{21} = 1$. Perform $R_2 \rightarrow R_2 - R_1$:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Step 2: Eliminate entries in Column 2 Target $A_{32} = 1$. Perform $R_3 \rightarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Step 3: Eliminate entries in Column 3 Target $A_{43} = 1$. Perform $R_4 \rightarrow R_4 - R_3$:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_4 - R_3} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final Echelon Form:

$$U = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Determinant Calculation: The determinant of an upper triangular matrix is the product of its diagonal entries.

$$\det(A) = 1 \times 1 \times 1 \times 1 = 1$$

Method 2: Tridiagonal Expansion Check

Instead of full row reduction, we can expand along the first row.

$$\det(A) = 1 \cdot \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Calculating the 3×3 minors:

1. Minor 1: $2(4 - 1) - 1(2 - 0) = 6 - 2 = 4$.
2. Minor 2: $1(4 - 1) - 1(0) = 3$.

$$\det(A) = 1(4) - 1(3) = 1$$

This confirms the result obtained via row reduction.

(b) Find Inverse and Solve $AX = b$ [2 Marks]

$$b = \begin{bmatrix} -2 \\ 2 \\ 1 \\ -1 \end{bmatrix}$$

Method 1: Explicit Inverse Calculation

We form the augmented matrix $[A|I]$ and row reduce until we get $[I|A^{-1}]$.

Start:

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array} \right]$$

Apply forward elimination (same as Part A):

$$\xrightarrow{\text{Forward Ops}} \left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array} \right]$$

Backward Elimination (Upwards): 1. $R_3 \rightarrow R_3 - R_4$:

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array} \right]$$

2. $R_2 \rightarrow R_2 - R_3$:

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -3 & 3 & -2 & 1 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array} \right]$$

3. $R_1 \rightarrow R_1 - R_2$:

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 4 & -3 & 2 & -1 \\ 0 & 1 & 0 & 0 & -3 & 3 & -2 & 1 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array} \right]$$

Final Inverse:

$$A^{-1} = \begin{bmatrix} 4 & -3 & 2 & -1 \\ -3 & 3 & -2 & 1 \\ 2 & -2 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

Solve $X = A^{-1}b$:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 & -3 & 2 & -1 \\ -3 & 3 & -2 & 1 \\ 2 & -2 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \\ 1 \\ -1 \end{bmatrix}$$

Calculation:

$$x_1 = 4(-2) - 3(2) + 2(1) - 1(-1) = -8 - 6 + 2 + 1 = -11$$

$$x_2 = -3(-2) + 3(2) - 2(1) + 1(-1) = 6 + 6 - 2 - 1 = 9$$

$$x_3 = 2(-2) - 2(2) + 2(1) - 1(-1) = -4 - 4 + 2 + 1 = -5$$

$$x_4 = -1(-2) + 1(2) - 1(1) + 1(-1) = 2 + 2 - 1 - 1 = 2$$

Method 2: Augmented System (Gauss Elimination)

Instead of finding the full inverse, we apply the row operations from Part (a) directly to the vector b .

Initial Vector: $b = [-2, 2, 1, -1]^T$.

1. Operation $R_2 \rightarrow R_2 - R_1$:

$$b'_2 = 2 - (-2) = 4$$

Vector becomes $[-2, 4, 1, -1]^T$.

2. Operation $R_3 \rightarrow R_3 - R_2$:

$$b'_3 = 1 - 4 = -3$$

Vector becomes $[-2, 4, -3, -1]^T$.

3. Operation $R_4 \rightarrow R_4 - R_3$:

$$b'_4 = -1 - (-3) = 2$$

Vector becomes $[-2, 4, -3, 2]^T$.

Solve Triangular System $UX = b'$:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ -3 \\ 2 \end{bmatrix}$$

Back Substitution Steps:

- Equation 4: $x_4 = 2$.
- Equation 3: $x_3 + x_4 = -3 \implies x_3 + 2 = -3 \implies x_3 = -5$.
- Equation 2: $x_2 + x_3 = 4 \implies x_2 + (-5) = 4 \implies x_2 = 9$.
- Equation 1: $x_1 + x_2 = -2 \implies x_1 + 9 = -2 \implies x_1 = -11$.

Final Solution: $X = [-11, 9, -5, 2]^T$.

2 Question 2: Subspaces

(a) Show that $V = \{X : AX = 0\}$ is a subspace [2 Marks]

$$A = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 2 & -2 \\ 3 & -3 & 3 \end{bmatrix}$$

Formal Proof

Let V be the null space of A . We check the three subspace axioms:

1. **Contains Zero Vector:**

$$A\mathbf{0} = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 2 & -2 \\ 3 & -3 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since $A\mathbf{0} = \mathbf{0}$, $\mathbf{0} \in V$.

2. **Closure under Addition:** Let $u, v \in V$. This implies $Au = \mathbf{0}$ and $Av = \mathbf{0}$.

$$A(u + v) = Au + Av = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

Since $A(u + v) = \mathbf{0}$, $u + v \in V$.

3. **Closure under Scalar Multiplication:** Let $u \in V$ and $c \in \mathbb{R}$.

$$A(cu) = c(Au) = c(\mathbf{0}) = \mathbf{0}$$

Since $A(cu) = \mathbf{0}$, $cu \in V$.

Therefore, V is a subspace of $\mathbb{R}^3$.

(b) Determine a Basis and Dimension [2 Marks]

Solution Steps

Step 1: Reduce Matrix to Echelon Form

$$\begin{bmatrix} 1 & -1 & 1 \\ -2 & 2 & -2 \\ 3 & -3 & 3 \end{bmatrix}$$

- $R_2 \rightarrow R_2 + 2R_1$: $[-2, 2, -2] + [2, -2, 2] = [0, 0, 0]$.
- $R_3 \rightarrow R_3 - 3R_1$: $[3, -3, 3] + [-3, 3, -3] = [0, 0, 0]$.

Reduced Row Echelon Form:

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 2: Solve for General Solution The pivot is in column 1. The free variables are corresponding to columns 2 and 3 (x_2 and x_3). Equation: $x_1 - x_2 + x_3 = 0 \implies x_1 =$

$x_2 - x_3$. Let $x_2 = s$ and $x_3 = t$.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} s - t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Step 3: Identify Basis and Dimension The vectors attached to the parameters s and t form the basis.

$$\text{Basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Since there are 2 vectors in the basis, **Dimension = 2**.

(c) Find a 1-dimensional subspace U [2 Marks]

Solution

A 1-dimensional subspace is spanned by a single non-zero vector. We can choose any vector from the basis of V found in part (b). Let us choose the first basis vector:

$$u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

The subspace U is defined as the span of u :

$$U = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Basis for U : $\{[1, 1, 0]^T\}$.

3 Question 3: Singular Value Decomposition

Problem: Find the SVD of $A = \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix}$.

Method 1: The Standard $A^T A$ Approach

Step 1: Compute $A^T A$

$$\begin{aligned} A^T A &= \begin{bmatrix} 5 & 5 & 0 \\ 0 & 0 & -10 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix} \\ &= \begin{bmatrix} (5)(5) + (5)(5) + 0 & 0 + 0 + 0 \\ 0 + 0 + 0 & 0 + 0 + (-10)(-10) \end{bmatrix} = \begin{bmatrix} 50 & 0 \\ 0 & 100 \end{bmatrix} \end{aligned}$$

Step 2: Find Singular Values The eigenvalues of the diagonal matrix are the diagonal entries.

$$\lambda_1 = 100, \quad \lambda_2 = 50$$

The singular values are $\sigma = \sqrt{\lambda}$.

$$\sigma_1 = \sqrt{100} = 10, \quad \sigma_2 = \sqrt{50} = 5\sqrt{2}$$

The matrix Σ (same dimensions as A , 3×2) is:

$$\Sigma = \begin{bmatrix} 10 & 0 \\ 0 & 5\sqrt{2} \\ 0 & 0 \end{bmatrix}$$

Step 3: Find Right Singular Vectors (V) We find normalized eigenvectors of $A^T A$.

- For $\lambda_1 = 100$:

$$(A^T A - 100I)x = \begin{bmatrix} -50 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies -50x_1 = 0 \implies x_1 = 0$$

Vector is $[0, 1]^T$. So $v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

- For $\lambda_2 = 50$:

$$(A^T A - 50I)x = \begin{bmatrix} 0 & 0 \\ 0 & 50 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies 50x_2 = 0 \implies x_2 = 0$$

Vector is $[1, 0]^T$. So $v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Step 4: Find Left Singular Vectors (U) We compute $u_i = \frac{1}{\sigma_i} A v_i$.

$$\bullet \quad u_1 = \frac{1}{10} \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 0 \\ 0 \\ -10 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}.$$

$$\bullet u_2 = \frac{1}{5\sqrt{2}} \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{5\sqrt{2}} \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}.$$

• u_3 : Needs to be orthogonal to u_1 and u_2 . Cross product $u_1 \times u_2 =$

$$\det \begin{bmatrix} i & j & k \\ 0 & 0 & -1 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} = i(1/\sqrt{2}) - j(1/\sqrt{2}) + k(0). \quad u_3 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}.$$

$$U = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ -1 & 0 & 0 \end{bmatrix}$$

Method 2: Column Orthogonality Check

Check the dot product of the columns of A :

$$C_1 \cdot C_2 = \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ -10 \end{bmatrix} = 0$$

Since columns are orthogonal, we can normalize them directly to get U .

- Length of C_2 is 10. This gives $\sigma_1 = 10$ and $u_1 = [0, 0, -1]^T$.
- Length of C_1 is $\sqrt{50}$. This gives $\sigma_2 = 5\sqrt{2}$ and $u_2 = [1/\sqrt{2}, 1/\sqrt{2}, 0]^T$.

This matches the result from Method 1 instantly.

4 Question 4: Symmetric Positive Definite Matrices

(a) Prove that A is Symmetric and Positive Definite [2 Marks]

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Proof

1. Symmetry: We verify that $A = A^T$. $A_{21} = 1 = A_{12}$, $A_{43} = 1 = A_{34}$. All other off-diagonal entries are 0. Thus, A is symmetric.

2. Positive Definiteness (Sylvester's Criterion): We calculate all leading principal minors (D_k).

- $D_1 = \det[2] = 2 > 0$.
- $D_2 = \det \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = (2)(3) - (1)(1) = 6 - 1 = 5 > 0$.
- $D_3 = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$. Expanding along row 3: $4 \times D_2 = 4 \times 5 = 20 > 0$.
- $D_4 = \det(A)$. Since A is block diagonal: $\det(A) = \det(\text{Block}_1) \times \det(\text{Block}_2) = 5 \times \det \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$. $\det(\text{Block}_2) = 8 - 1 = 7$. $D_4 = 5 \times 7 = 35 > 0$.

Since all leading principal minors are positive, A is Positive Definite.

(b) Compute the distance between vectors [2 Marks]

$$u = \begin{bmatrix} -2 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \quad v = \begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

Calculation

Step 1: Compute Difference Vector $w = u - v$

$$w = \begin{bmatrix} -2 - (-2) \\ 0 - 2 \\ 1 - 1 \\ -1 - 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \\ -1 \end{bmatrix}$$

Step 2: Compute Norm Squared $\|w\|_A^2 = w^T A w$ First, compute Aw :

$$Aw = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ -2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1(-2) \\ 3(-2) \\ 1(-1) \\ 2(-1) \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \\ -1 \\ -2 \end{bmatrix}$$

Now compute the dot product $w^T(Aw)$:

$$\begin{aligned} & \begin{bmatrix} 0 & -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ -6 \\ -1 \\ -2 \end{bmatrix} \\ &= 0(-2) + (-2)(-6) + 0(-1) + (-1)(-2) \\ &= 0 + 12 + 0 + 2 = 14 \end{aligned}$$

Step 3: Final Distance

$$\text{Distance} = \sqrt{14}$$

5 Question 5: Calculus

(a) Find the gradient of $f(x, y, z)$ [2 Marks]

$$f(x, y, z) = xy^2 + yz^2 + zx^2$$

Solution

We calculate the partial derivatives:

$$\frac{\partial f}{\partial x} = y^2(1) + 0 + z(2x) = y^2 + 2zx$$

$$\frac{\partial f}{\partial y} = x(2y) + z^2(1) + 0 = 2xy + z^2$$

$$\frac{\partial f}{\partial z} = 0 + y(2z) + x^2(1) = 2yz + x^2$$

Gradient Vector:

$$\nabla f = \begin{bmatrix} y^2 + 2zx \\ 2xy + z^2 \\ 2yz + x^2 \end{bmatrix}$$

(b) Find critical points of $g(x, y) = f(x, y, 0)$ [1 Mark]

Solution

Substitute $z = 0$ into f :

$$g(x, y) = xy^2 + y(0)^2 + (0)x^2 = xy^2$$

Calculate Gradient of g :

$$\nabla g = \begin{bmatrix} \frac{\partial g}{\partial x} \\ \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} y^2 \\ 2xy \end{bmatrix}$$

Set $\nabla g = \mathbf{0}$:

$$1. \ y^2 = 0 \implies y = 0.$$

$$2. \ 2xy = 0 \implies 2x(0) = 0, \text{ which is } 0 = 0.$$

Since the second equation is satisfied for any x , the critical points are all points where $y = 0$. **Critical Points:** The set $\{(x, 0) \mid x \in \mathbb{R}\}$ (the entire x-axis).

(c) Examine Hessian Test [2 Marks]

Analysis

Compute the Hessian Matrix of $g(x, y) = xy^2$.

$$g_{xx} = 0, \quad g_{xy} = 2y, \quad g_{yx} = 2y, \quad g_{yy} = 2x$$

$$H = \begin{bmatrix} 0 & 2y \\ 2y & 2x \end{bmatrix}$$

Evaluate H at the critical points $(x, 0)$ (where $y = 0$):

$$H|_{(x,0)} = \begin{bmatrix} 0 & 0 \\ 0 & 2x \end{bmatrix}$$

Calculate the Determinant:

$$\det(H) = (0)(2x) - (0)(0) = 0$$

Conclusion: Since the determinant of the Hessian is zero, the Second Derivative Test is **inconclusive** (it fails). We cannot classify the points using this method.

6 Question 6: Taylor Series

(a) Find Taylor polynomial of degree 2 [4 Marks]

$$f(x) = \exp(\sin(x^2)) \quad \text{at } x = 0$$

Method 1: Direct Substitution (Recommended)

Using standard Taylor series expansions around 0:

1. $\sin(u) \approx u - \frac{u^3}{6} \dots$ Let $u = x^2$. Then $\sin(x^2) \approx x^2$. (Higher terms are degree 6, which we ignore).

2. $e^t \approx 1 + t + \frac{t^2}{2} \dots$ Let $t = \sin(x^2) \approx x^2$.

Substitute $t \approx x^2$ into the exponential series:

$$e^{\sin(x^2)} \approx 1 + (x^2) + \frac{(x^2)^2}{2}$$

Keeping terms only up to degree 2:

$$P_2(x) = 1 + x^2$$

Method 2: Calculus Derivatives

1. $f(0) = e^{\sin(0)} = e^0 = 1$. 2. First Derivative:

$$f'(x) = e^{\sin(x^2)} \cdot \cos(x^2) \cdot 2x$$

$$f'(0) = 1 \cdot 1 \cdot 0 = 0$$

3. Second Derivative (Product Rule):

$$f''(x) = \frac{d}{dx} [e^{\sin(x^2)} \cos(x^2)(2x)]$$

At $x = 0$, terms with $2x$ will vanish because $2(0) = 0$. We only care about the derivative of the $(2x)$ term:

$$f''(0) = [e^{\sin 0} \cos 0] \cdot \frac{d}{dx}(2x) = 1 \cdot 1 \cdot 2 = 2$$

4. Construct Polynomial:

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$$

$$P_2(x) = 1 + 0x + \frac{2}{2}x^2 = 1 + x^2$$

(b) Smallest and Largest Values [1 Mark]

Solution

Consider the inner function $u = \sin(x^2)$. The range of $\sin(\theta)$ is always $[-1, 1]$. The function $f(u) = e^u$ is a strictly increasing function. Therefore:

- **Smallest Value:** Occurs when $\sin(x^2) = -1$. Value $= e^{-1} = 1/e$.

- **Largest Value:** Occurs when $\sin(x^2) = 1$. Value = $e^1 = e$.